GV-6.2-003	Establish incident response plans for third-party GAI technologies: Align incident response plans with impacts enumerated in MAP 5.1; Communicate third-party GAI incident response plans to all relevant AI Actors; Define ownership of GAI incident response functions; Rehearse third-party GAI incident response plans at a regular cadence; Improve incident response plans based on retrospective learning; Review incident response plans for alignment with relevant breach reporting, data protection, data privacy, or other laws.	Data Privacy; Human-AI Configuration; Information Security; Value Chain and Component Integration; Harmful Bias and Homogenization
GV-6.2-004	Establish policies and procedures for continuous monitoring of third-party GAI systems in deployment.	Value Chain and Component Integration
GV-6.2-005	Establish policies and procedures that address GAI data redundancy, including model weights and other system artifacts.	Harmful Bias and Homogenization
GV-6.2-006	Establish policies and procedures to test and manage risks related to rollover and fallback technologies for GAI systems, acknowledging that rollover and fallback may include manual processing.	Information Integrity
GV-6.2-007	Review vendor contracts and avoid arbitrary or capricious termination of critical GAI technologies or vendor services and non-standard terms that may amplify or defer liability in unexpected ways and/or contribute to unauthorized data collection by vendors or third-parties (e.g., secondary data use). Consider: Clear assignment of liability and responsibility for incidents, GAI system changes over time (e.g., fine-tuning, drift, decay); Request: Notification and disclosure for serious incidents arising from third-party data and systems; Service Level Agreements (SLAs) in vendor contracts that address incident response, response times, and availability of critical support.	Human-AI Configuration; Information Security; Value Chain and Component Integration
AI Actor Tasks: AI Deployment, Operation and Monitoring, TEVV, Third-party entities		

MAP 1.1: Intended purposes, potentially beneficial uses, context specific laws, norms and expectations, and prospective settings in which the AI system will be deployed are understood and documented. Considerations include: the specific set or types of users along with their expectations; potential positive and negative impacts of system uses to individuals, communities, organizations, society, and the planet; assumptions and related limitations about AI system purposes, uses, and risks across the development or product AI lifecycle; and related TEVV and system metrics.

Action ID	**Suggested Action**	**GAI Risks**
MP-1.1-001	When identifying intended purposes, consider factors such as internal vs. external use, narrow vs. broad application scope, fine-tuning, and varieties of data sources (e.g., grounding, retrieval-augmented generation).	Data Privacy; Intellectual Property

MP-1.1-002	Determine and document the expected and acceptable GAI system context of use in collaboration with socio-cultural and other domain experts, by assessing: Assumptions and limitations; Direct value to the organization; Intended operational environment and observed usage patterns; Potential positive and negative impacts to individuals, public safety, groups, communities, organizations, democratic institutions, and the physical environment; Social norms and expectations.	Harmful Bias and Homogenization
MP-1.1-003	Document risk measurement plans to address identified risks. Plans may include, as applicable: Individual and group cognitive biases (e.g., confirmation bias, funding bias, groupthink) for AI Actors involved in the design, implementation, and use of GAI systems; Known past GAI system incidents and failure modes; In-context use and foreseeable misuse, abuse, and off-label use; Over reliance on quantitative metrics and methodologies without sufficient awareness of their limitations in the context(s) of use; Standard measurement and structured human feedback approaches; Anticipated human-AI configurations.	Human-AI Configuration; Harmful Bias and Homogenization; Dangerous, Violent, or Hateful Content
MP-1.1-004	Identify and document foreseeable illegal uses or applications of the GAI system that surpass organizational risk tolerances.	CBRN Information or Capabilities; Dangerous, Violent, or Hateful Content; Obscene, Degrading, and/or Abusive Content
AI Actor Tasks: AI Deployment		

MAP 1.2: Interdisciplinary AI Actors, competencies, skills, and capacities for establishing context reflect demographic diversity and broad domain and user experience expertise, and their participation is documented. Opportunities for interdisciplinary collaboration are prioritized.		
Action ID	**Suggested Action**	**GAI Risks**
MP-1.2-001	Establish and empower interdisciplinary teams that reflect a wide range of capabilities, competencies, demographic groups, domain expertise, educational backgrounds, lived experiences, professions, and skills across the enterprise to inform and conduct risk measurement and management functions.	Human-AI Configuration; Harmful Bias and Homogenization
MP-1.2-002	Verify that data or benchmarks used in risk measurement, and users, participants, or subjects involved in structured GAI public feedback exercises are representative of diverse in-context user populations.	Human-AI Configuration; Harmful Bias and Homogenization
AI Actor Tasks: AI Deployment		

MAP 2.1: The specific tasks and methods used to implement the tasks that the AI system will support are defined (e.g., classifiers, generative models, recommenders).		
Action ID	**Suggested Action**	**GAI Risks**
MP-2.1-001	Establish known assumptions and practices for determining data origin and content lineage, for documentation and evaluation purposes.	Information Integrity
MP-2.1-002	Institute test and evaluation for data and content flows within the GAI system, including but not limited to, original data sources, data transformations, and decision-making criteria.	Intellectual Property; Data Privacy
AI Actor Tasks: TEVV		

MAP 2.2: Information about the AI system's knowledge limits and how system output may be utilized and overseen by humans is documented. Documentation provides sufficient information to assist relevant AI Actors when making decisions and taking subsequent actions.		
Action ID	**Suggested Action**	**GAI Risks**
MP-2.2-001	Identify and document how the system relies on upstream data sources, including for content provenance, and if it serves as an upstream dependency for other systems.	Information Integrity; Value Chain and Component Integration
MP-2.2-002	Observe and analyze how the GAI system interacts with external networks, and identify any potential for negative externalities, particularly where content provenance might be compromised.	Information Integrity
AI Actor Tasks: End Users		

MAP 2.3: Scientific integrity and TEVV considerations are identified and documented, including those related to experimental design, data collection and selection (e.g., availability, representativeness, suitability), system trustworthiness, and construct validation		
Action ID	**Suggested Action**	**GAI Risks**
MP-2.3-001	Assess the accuracy, quality, reliability, and authenticity of GAI output by comparing it to a set of known ground truth data and by using a variety of evaluation methods (e.g., human oversight and automated evaluation, proven cryptographic techniques, review of content inputs).	Information Integrity

MP-2.3-002	Review and document accuracy, representativeness, relevance, suitability of data used at different stages of AI life cycle.	Harmful Bias and Homogenization; Intellectual Property
MP-2.3-003	Deploy and document fact-checking techniques to verify the accuracy and veracity of information generated by GAI systems, especially when the information comes from multiple (or unknown) sources.	Information Integrity
MP-2.3-004	Develop and implement testing techniques to identify GAI produced content (e.g., synthetic media) that might be indistinguishable from human-generated content.	Information Integrity
MP-2.3-005	Implement plans for GAI systems to undergo regular adversarial testing to identify vulnerabilities and potential manipulation or misuse.	Information Security
AI Actor Tasks: AI Development, Domain Experts, TEVV		

MAP 3.4: Processes for operator and practitioner proficiency with AI system performance and trustworthiness – and relevant technical standards and certifications – are defined, assessed, and documented.		
Action ID	**Suggested Action**	**GAI Risks**
MP-3.4-001	Evaluate whether GAI operators and end-users can accurately understand content lineage and origin.	Human-AI Configuration; Information Integrity
MP-3.4-002	Adapt existing training programs to include modules on digital content transparency.	Information Integrity
MP-3.4-003	Develop certification programs that test proficiency in managing GAI risks and interpreting content provenance, relevant to specific industry and context.	Information Integrity
MP-3.4-004	Delineate human proficiency tests from tests of GAI capabilities.	Human-AI Configuration
MP-3.4-005	Implement systems to continually monitor and track the outcomes of human-GAI configurations for future refinement and improvements.	Human-AI Configuration; Information Integrity
MP-3.4-006	Involve the end-users, practitioners, and operators in GAI system in prototyping and testing activities. Make sure these tests cover various scenarios, such as crisis situations or ethically sensitive contexts.	Human-AI Configuration; Information Integrity; Harmful Bias and Homogenization; Dangerous, Violent, or Hateful Content
AI Actor Tasks: AI Design, AI Development, Domain Experts, End-Users, Human Factors, Operation and Monitoring		

MAP 4.1: Approaches for mapping AI technology and legal risks of its components – including the use of third-party data or software – are in place, followed, and documented, as are risks of infringement of a third-party's intellectual property or other rights.		
Action ID	**Suggested Action**	**GAI Risks**
MP-4.1-001	Conduct periodic monitoring of AI-generated content for privacy risks; address any possible instances of PII or sensitive data exposure.	Data Privacy
MP-4.1-002	Implement processes for responding to potential intellectual property infringement claims or other rights.	Intellectual Property
MP-4.1-003	Connect new GAI policies, procedures, and processes to existing model, data, software development, and IT governance and to legal, compliance, and risk management activities.	Information Security; Data Privacy
MP-4.1-004	Document training data curation policies, to the extent possible and according to applicable laws and policies.	Intellectual Property; Data Privacy; Obscene, Degrading, and/or Abusive Content
MP-4.1-005	Establish policies for collection, retention, and minimum quality of data, in consideration of the following risks: Disclosure of inappropriate CBRN information; Use of Illegal or dangerous content; Offensive cyber capabilities; Training data imbalances that could give rise to harmful biases; Leak of personally identifiable information, including facial likenesses of individuals.	CBRN Information or Capabilities; Intellectual Property; Information Security; Harmful Bias and Homogenization; Dangerous, Violent, or Hateful Content; Data Privacy
MP-4.1-006	Implement policies and practices defining how third-party intellectual property and training data will be used, stored, and protected.	Intellectual Property; Value Chain and Component Integration
MP-4.1-007	Re-evaluate models that were fine-tuned or enhanced on top of third-party models.	Value Chain and Component Integration
MP-4.1-008	Re-evaluate risks when adapting GAI models to new domains. Additionally, establish warning systems to determine if a GAI system is being used in a new domain where previous assumptions (relating to context of use or mapped risks such as security, and safety) may no longer hold.	CBRN Information or Capabilities; Intellectual Property; Harmful Bias and Homogenization; Dangerous, Violent, or Hateful Content; Data Privacy
MP-4.1-009	Leverage approaches to detect the presence of PII or sensitive data in generated output text, image, video, or audio.	Data Privacy

MP-4.1-010	Conduct appropriate diligence on training data use to assess intellectual property, and privacy, risks, including to examine whether use of proprietary or sensitive training data is consistent with applicable laws.	Intellectual Property; Data Privacy
AI Actor Tasks: Governance and Oversight, Operation and Monitoring, Procurement, Third-party entities		

MAP 5.1: Likelihood and magnitude of each identified impact (both potentially beneficial and harmful) based on expected use, past uses of AI systems in similar contexts, public incident reports, feedback from those external to the team that developed or deployed the AI system, or other data are identified and documented.		
Action ID	**Suggested Action**	**GAI Risks**
MP-5.1-001	Apply TEVV practices for content provenance (e.g., probing a system's synthetic data generation capabilities for potential misuse or vulnerabilities.	Information Integrity; Information Security
MP-5.1-002	Identify potential content provenance harms of GAI, such as misinformation or disinformation, deepfakes, including NCII, or tampered content. Enumerate and rank risks based on their likelihood and potential impact, and determine how well provenance solutions address specific risks and/or harms.	Information Integrity; Dangerous, Violent, or Hateful Content; Obscene, Degrading, and/or Abusive Content
MP-5.1-003	Consider disclosing use of GAI to end users in relevant contexts, while considering the objective of disclosure, the context of use, the likelihood and magnitude of the risk posed, the audience of the disclosure, as well as the frequency of the disclosures.	Human-AI Configuration
MP-5.1-004	Prioritize GAI structured public feedback processes based on risk assessment estimates.	Information Integrity; CBRN Information or Capabilities; Dangerous, Violent, or Hateful Content; Harmful Bias and Homogenization
MP-5.1-005	Conduct adversarial role-playing exercises, GAI red-teaming, or chaos testing to identify anomalous or unforeseen failure modes.	Information Security
MP-5.1-006	Profile threats and negative impacts arising from GAI systems interacting with, manipulating, or generating content, and outlining known and potential vulnerabilities and the likelihood of their occurrence.	Information Security
AI Actor Tasks: AI Deployment, AI Design, AI Development, AI Impact Assessment, Affected Individuals and Communities, End-Users, Operation and Monitoring		

MAP 5.2: Practices and personnel for supporting regular engagement with relevant AI Actors and integrating feedback about positive, negative, and unanticipated impacts are in place and documented.		
Action ID	**Suggested Action**	**GAI Risks**
MP-5.2-001	Determine context-based measures to identify if new impacts are present due to the GAI system, including regular engagements with downstream AI Actors to identify and quantify new contexts of unanticipated impacts of GAI systems.	Human-AI Configuration; Value Chain and Component Integration
MP-5.2-002	Plan regular engagements with AI Actors responsible for inputs to GAI systems, including third-party data and algorithms, to review and evaluate unanticipated impacts.	Human-AI Configuration; Value Chain and Component Integration
AI Actor Tasks: AI Deployment, AI Design, AI Impact Assessment, Affected Individuals and Communities, Domain Experts, End-Users, Human Factors, Operation and Monitoring		

MEASURE 1.1: Approaches and metrics for measurement of AI risks enumerated during the MAP function are selected for implementation starting with the most significant AI risks. The risks or trustworthiness characteristics that will not – or cannot – be measured are properly documented.		
Action ID	**Suggested Action**	**GAI Risks**
MS-1.1-001	Employ methods to trace the origin and modifications of digital content.	Information Integrity
MS-1.1-002	Integrate tools designed to analyze content provenance and detect data anomalies, verify the authenticity of digital signatures, and identify patterns associated with misinformation or manipulation.	Information Integrity
MS-1.1-003	Disaggregate evaluation metrics by demographic factors to identify any discrepancies in how content provenance mechanisms work across diverse populations.	Information Integrity; Harmful Bias and Homogenization
MS-1.1-004	Develop a suite of metrics to evaluate structured public feedback exercises informed by representative AI Actors.	Human-AI Configuration; Harmful Bias and Homogenization; CBRN Information or Capabilities
MS-1.1-005	Evaluate novel methods and technologies for the measurement of GAI-related risks including in content provenance, offensive cyber, and CBRN, while maintaining the models' ability to produce valid, reliable, and factually accurate outputs.	Information Integrity; CBRN Information or Capabilities; Obscene, Degrading, and/or Abusive Content

MS-1.1-006	Implement continuous monitoring of GAI system impacts to identify whether GAI outputs are equitable across various sub-populations. Seek active and direct feedback from affected communities via structured feedback mechanisms or red-teaming to monitor and improve outputs.	Harmful Bias and Homogenization
MS-1.1-007	Evaluate the quality and integrity of data used in training and the provenance of AI-generated content, for example by employing techniques like chaos engineering and seeking stakeholder feedback.	Information Integrity
MS-1.1-008	Define use cases, contexts of use, capabilities, and negative impacts where structured human feedback exercises, e.g., GAI red-teaming, would be most beneficial for GAI risk measurement and management based on the context of use.	Harmful Bias and Homogenization; CBRN Information or Capabilities
MS-1.1-009	Track and document risks or opportunities related to all GAI risks that cannot be measured quantitatively, including explanations as to why some risks cannot be measured (e.g., due to technological limitations, resource constraints, or trustworthy considerations). Include unmeasured risks in marginal risks.	Information Integrity
AI Actor Tasks: AI Development, Domain Experts, TEVV		

MEASURE 1.3: Internal experts who did not serve as front-line developers for the system and/or independent assessors are involved in regular assessments and updates. Domain experts, users, AI Actors external to the team that developed or deployed the AI system, and affected communities are consulted in support of assessments as necessary per organizational risk tolerance.		
Action ID	**Suggested Action**	**GAI Risks**
MS-1.3-001	Define relevant groups of interest (e.g., demographic groups, subject matter experts, experience with GAI technology) within the context of use as part of plans for gathering structured public feedback.	Human-AI Configuration; Harmful Bias and Homogenization; CBRN Information or Capabilities
MS-1.3-002	Engage in internal and external evaluations, GAI red-teaming, impact assessments, or other structured human feedback exercises in consultation with representative AI Actors with expertise and familiarity in the context of use, and/or who are representative of the populations associated with the context of use.	Human-AI Configuration; Harmful Bias and Homogenization; CBRN Information or Capabilities
MS-1.3-003	Verify those conducting structured human feedback exercises are not directly involved in system development tasks for the same GAI model.	Human-AI Configuration; Data Privacy
AI Actor Tasks: AI Deployment, AI Development, AI Impact Assessment, Affected Individuals and Communities, Domain Experts, End-Users, Operation and Monitoring, TEVV		

MEASURE 2.2: Evaluations involving human subjects meet applicable requirements (including human subject protection) and are representative of the relevant population.		
Action ID	**Suggested Action**	**GAI Risks**
MS-2.2-001	Assess and manage statistical biases related to GAI content provenance through techniques such as re-sampling, re-weighting, or adversarial training.	Information Integrity; Information Security; Harmful Bias and Homogenization
MS-2.2-002	Document how content provenance data is tracked and how that data interacts with privacy and security. Consider: Anonymizing data to protect the privacy of human subjects; Leveraging privacy output filters; Removing any personally identifiable information (PII) to prevent potential harm or misuse.	Data Privacy; Human AI Configuration; Information Integrity; Information Security; Dangerous, Violent, or Hateful Content
MS-2.2-003	Provide human subjects with options to withdraw participation or revoke their consent for present or future use of their data in GAI applications.	Data Privacy; Human-AI Configuration; Information Integrity
MS-2.2-004	Use techniques such as anonymization, differential privacy or other privacy-enhancing technologies to minimize the risks associated with linking AI-generated content back to individual human subjects.	Data Privacy; Human-AI Configuration
AI Actor Tasks: AI Development, Human Factors, TEVV		

MEASURE 2.3: AI system performance or assurance criteria are measured qualitatively or quantitatively and demonstrated for conditions similar to deployment setting(s). Measures are documented.		
Action ID	**Suggested Action**	**GAI Risks**
MS-2.3-001	Consider baseline model performance on suites of benchmarks when selecting a model for fine tuning or enhancement with retrieval-augmented generation.	Information Security; Confabulation
MS-2.3-002	Evaluate claims of model capabilities using empirically validated methods.	Confabulation; Information Security
MS-2.3-003	Share results of pre-deployment testing with relevant GAI Actors, such as those with system release approval authority.	Human-AI Configuration

MS-2.3-004	Utilize a purpose-built testing environment such as NIST Dioptra to empirically evaluate GAI trustworthy characteristics.	CBRN Information or Capabilities; Data Privacy; Confabulation; Information Integrity; Information Security; Dangerous, Violent, or Hateful Content; Harmful Bias and Homogenization
AI Actor Tasks: AI Deployment, TEVV		

MEASURE 2.5: The AI system to be deployed is demonstrated to be valid and reliable. Limitations of the generalizability beyond the conditions under which the technology was developed are documented.		
Action ID	**Suggested Action**	**Risks**
MS-2.5-001	Avoid extrapolating GAI system performance or capabilities from narrow, non-systematic, and anecdotal assessments.	Human-AI Configuration; Confabulation
MS-2.5-002	Document the extent to which human domain knowledge is employed to improve GAI system performance, via, e.g., RLHF, fine-tuning, retrieval-augmented generation, content moderation, business rules.	Human-AI Configuration
MS-2.5-003	Review and verify sources and citations in GAI system outputs during pre-deployment risk measurement and ongoing monitoring activities.	Confabulation
MS-2.5-004	Track and document instances of anthropomorphization (e.g., human images, mentions of human feelings, cyborg imagery or motifs) in GAI system interfaces.	Human-AI Configuration
MS-2.5-005	Verify GAI system training data and TEVV data provenance, and that fine-tuning or retrieval-augmented generation data is grounded.	Information Integrity
MS-2.5-006	Regularly review security and safety guardrails, especially if the GAI system is being operated in novel circumstances. This includes reviewing reasons why the GAI system was initially assessed as being safe to deploy.	Information Security; Dangerous, Violent, or Hateful Content
AI Actor Tasks: Domain Experts, TEVV		

MEASURE 2.6: The AI system is evaluated regularly for safety risks – as identified in the MAP function. The AI system to be deployed is demonstrated to be safe, its residual negative risk does not exceed the risk tolerance, and it can fail safely, particularly if made to operate beyond its knowledge limits. Safety metrics reflect system reliability and robustness, real-time monitoring, and response times for AI system failures.		
Action ID	**Suggested Action**	**GAI Risks**
MS-2.6-001	Assess adverse impacts, including health and wellbeing impacts for value chain or other AI Actors that are exposed to sexually explicit, offensive, or violent information during GAI training and maintenance.	Human-AI Configuration; Obscene, Degrading, and/or Abusive Content; Value Chain and Component Integration; Dangerous, Violent, or Hateful Content
MS-2.6-002	Assess existence or levels of harmful bias, intellectual property infringement, data privacy violations, obscenity, extremism, violence, or CBRN information in system training data.	Data Privacy; Intellectual Property; Obscene, Degrading, and/or Abusive Content; Harmful Bias and Homogenization; Dangerous, Violent, or Hateful Content; CBRN Information or Capabilities
MS-2.6-003	Re-evaluate safety features of fine-tuned models when the negative risk exceeds organizational risk tolerance.	Dangerous, Violent, or Hateful Content
MS-2.6-004	Review GAI system outputs for validity and safety: Review generated code to assess risks that may arise from unreliable downstream decision-making.	Value Chain and Component Integration; Dangerous, Violent, or Hateful Content
MS-2.6-005	Verify that GAI system architecture can monitor outputs and performance, and handle, recover from, and repair errors when security anomalies, threats and impacts are detected.	Confabulation; Information Integrity; Information Security
MS-2.6-006	Verify that systems properly handle queries that may give rise to inappropriate, malicious, or illegal usage, including facilitating manipulation, extortion, targeted impersonation, cyber-attacks, and weapons creation.	CBRN Information or Capabilities; Information Security
MS-2.6-007	Regularly evaluate GAI system vulnerabilities to possible circumvention of safety measures.	CBRN Information or Capabilities; Information Security
AI Actor Tasks: AI Deployment, AI Impact Assessment, Domain Experts, Operation and Monitoring, TEVV		

MEASURE 2.7: AI system security and resilience – as identified in the MAP function – are evaluated and documented.		
Action ID	**Suggested Action**	**GAI Risks**
MS-2.7-001	Apply established security measures to: Assess likelihood and magnitude of vulnerabilities and threats such as backdoors, compromised dependencies, data breaches, eavesdropping, man-in-the-middle attacks, reverse engineering, autonomous agents, model theft or exposure of model weights, AI inference, bypass, extraction, and other baseline security concerns.	Data Privacy; Information Integrity; Information Security; Value Chain and Component Integration
MS-2.7-002	Benchmark GAI system security and resilience related to content provenance against industry standards and best practices. Compare GAI system security features and content provenance methods against industry state-of-the-art.	Information Integrity; Information Security
MS-2.7-003	Conduct user surveys to gather user satisfaction with the AI-generated content and user perceptions of content authenticity. Analyze user feedback to identify concerns and/or current literacy levels related to content provenance and understanding of labels on content.	Human-AI Configuration; Information Integrity
MS-2.7-004	Identify metrics that reflect the effectiveness of security measures, such as data provenance, the number of unauthorized access attempts, inference, bypass, extraction, penetrations, or provenance verification.	Information Integrity; Information Security
MS-2.7-005	Measure reliability of content authentication methods, such as watermarking, cryptographic signatures, digital fingerprints, as well as access controls, conformity assessment, and model integrity verification, which can help support the effective implementation of content provenance techniques. Evaluate the rate of false positives and false negatives in content provenance, as well as true positives and true negatives for verification.	Information Integrity
MS-2.7-006	Measure the rate at which recommendations from security checks and incidents are implemented. Assess how quickly the AI system can adapt and improve based on lessons learned from security incidents and feedback.	Information Integrity; Information Security
MS-2.7-007	Perform AI red-teaming to assess resilience against: Abuse to facilitate attacks on other systems (e.g., malicious code generation, enhanced phishing content), GAI attacks (e.g., prompt injection), ML attacks (e.g., adversarial examples/prompts, data poisoning, membership inference, model extraction, sponge examples).	Information Security; Harmful Bias and Homogenization; Dangerous, Violent, or Hateful Content
MS-2.7-008	Verify fine-tuning does not compromise safety and security controls.	Information Integrity; Information Security; Dangerous, Violent, or Hateful Content